

ECC006 — Web Design & Development

Homework Assignment #7

HTML Fragments • CSS Pseudo-elements • Mobile Design • Error Correction

Part 1: Multiple Choice Questions

For each question, the correct answer is highlighted in green with a checkmark (✓). An explanation is provided below each question.

Question 1: How would you link to the named fragment #jobs on the page employ.html from the home page of the site?

- ✓ a. `<a href="employ.html#jobs">Jobs</a>`
- b. `<a name="employ.html#jobs">Jobs</a>`
- c. `<a link="employ.html#jobs">Jobs</a>`
- d. `<a href="#jobs">Jobs</a>`

Explanation: To link to a named fragment on a different page, the href attribute must include the filename followed by # and the fragment name. Option d. uses only #jobs, which only works on the same page. Options b. and c. use invalid attributes (name and link are not used this way in anchor tags).

Question 2: Which pseudo-element can be used to generate content that precedes an element?

- a. `:after`
- ✓ b. `:before`
- c. `:content`
- d. `:first-line`

Explanation: The CSS `:before` pseudo-element inserts generated content immediately before the selected element's content. The `:after` pseudo-element inserts content after the element. The `:content` and `:first-line` pseudo-elements serve different purposes and cannot generate preceding content.

Question 3: Which of the following is a mobile web design best practice?

- a. Configure a multiple-column page layout.
- b. Avoid using lists to organize information.
- ✓ c. Configure a single-column page layout.

d. Embed text in images wherever possible.

Explanation: Single-column layouts are recommended for mobile devices because small screens cannot comfortably display multiple columns side by side. Multiple columns require horizontal scrolling or zooming. Embedding text in images is discouraged as it cannot be scaled properly and is inaccessible. Lists are actually recommended for organizing information on mobile devices.

Part 2: Find the Error (Question 4)

The page below is intended for the navigation area to display on the right side of the browser window. Two changes must be made to the CSS:

✖ Original Code (Incorrect)	✔ Corrected Code
<pre>nav { float: left; width: 120px; } main { padding: 20px 150px 20px 20px; ... }</pre>	<pre>nav { float: right; width: 120px; } main { padding: 20px 20px 20px 150px; ... }</pre>

Explanation of Changes

- **Change 1 — float: left → float: right:** The nav element was set to float: left, which placed the navigation on the left side of the page. Changing this to float: right moves the navigation bar to the right side as intended.
- **Change 2 — Padding adjustment on main:** The original padding (20px 150px 20px 20px) reserved 150px of space on the right side. Since the nav is now on the right, the large padding value must move to the left side. The corrected value (20px 20px 20px 150px) reserves 150px on the left, preventing the main content from overlapping the navigation.

Part 3: Web Research Summary

Accessibility and Mobile Web Design: Areas of Overlap and Shared Best Practices

The W3C document Web Content Accessibility and Mobile Web, published at <http://www.w3.org/WAI/mobile>, demonstrates that designing for users with disabilities and designing for mobile devices are not separate tasks — they share a significant number of common goals, constraints, and technical solutions. Web developers who understand this overlap can build websites that serve a broader audience more effectively with less duplicated effort.

1. Alternative Text for Images

Both accessibility guidelines and mobile best practices require the use of descriptive alt attributes on images. Screen reader software used by visually impaired users depends entirely on alt text to convey the meaning of images. Similarly, mobile users on limited data connections often disable image loading to reduce bandwidth consumption — in those cases, the alt text becomes the only information they receive. Providing meaningful alt attributes therefore satisfies both audiences simultaneously.

2. Logical Heading Structure

Proper use of HTML heading elements (h1 through h6) benefits both groups of users. Assistive technologies like screen readers allow users to navigate a page by jumping between headings, making a clear hierarchy essential for accessibility. On small mobile screens, well-structured headings also allow users to quickly scan and understand a page's organization without reading every word. Developers should use headings semantically — not for styling purposes — to satisfy both requirements.

3. Single-Column Layouts

The W3C guidance notes that complex multi-column layouts create problems for screen reader users, who hear content in the order it appears in the HTML rather than visually on screen. The same layouts are also problematic on narrow mobile screens, where horizontal scrolling becomes necessary. A single-column, linearly structured layout eliminates both issues at once, making content readable and logically ordered for screen readers while remaining visually clear on small displays.

4. Avoiding Text Embedded in Images

Embedding text within images is discouraged by both accessibility and mobile guidelines for similar reasons. Screen readers cannot extract or read text that is part of an image file. On mobile devices, image-based text does not scale properly when users increase their browser font size or when the display resolution changes, resulting in blurry or unreadable content. Using real HTML text with CSS styling instead of text in images solves both problems.

5. Keyboard and Touch Navigation

Accessible websites must be fully operable using only a keyboard, which benefits users with motor disabilities who cannot use a mouse. This requirement aligns closely with mobile touch navigation, where interactive elements must be large enough and spaced adequately to be tapped accurately with a finger. Both standards push developers toward clear, well-spaced interactive controls — buttons, links, and form fields — that are easy to activate regardless of input method.

6. Concise and Clearly Organized Content

Screen reader users experience web content one element at a time, making dense or poorly organized text frustrating and difficult to navigate. Mobile users similarly benefit from concise, well-organized content that does not require extensive scrolling or zooming. Both audiences are best served by short paragraphs, descriptive link text, and the use of lists to group related information rather than long unbroken blocks of prose.

Conclusion

The overlap between web accessibility and mobile design best practices is not coincidental — both fields address the challenge of delivering content clearly under constrained conditions. Developers who adopt an accessibility-first mindset when building websites will naturally create products that also perform well on mobile devices. Conversely, mobile-first development encourages the structural simplicity that benefits users of assistive technologies. Treating accessibility and mobile compatibility as a unified set of requirements rather than two separate concerns leads to more inclusive, maintainable, and effective web design.